

A32

$$F(z) = F(X+Y) = F_X(X)F_Y(Y) =$$

|              |              |              |
|--------------|--------------|--------------|
| <del>0</del> | <del>z</del> | <del>0</del> |
| <del>1</del> | <del>p</del> |              |

A32

~~$$F(z) = P(X+Y \leq z) =$$~~

$$P(Z=1) = P(X=0, Y=1) = 0.2 \times 0.2 = 0.04$$

$$P(Z=2) = P(X=1, Y=1) + P(X=0, Y=2) = 0.06 + 0.04 = 0.14$$

$$P(Z=3) = P(X=2, Y=1) + P(X=1, Y=2) + P(X=0, Y=3) = 0.1 + 0.12 + 0.08 = 0.3$$

$$P(Z=4) = P(X=1, Y=2) + P(X=2, Y=2) = 0.12 + 0.2 = 0.32$$

$$P(Z=5) = P(X=2, Y=3) = 0.2$$

z 1 2 3 4 5

P 0.04 0.14 0.3 0.32 0.2

~~$$P(M=1) = P(X=1, Y=1) = 0$$~~

~~$$P(M=1) = P(X \leq 1, Y \leq 1) = 0.5 \times 0.2 = 0.1$$~~

~~$$P(M=2) = P(M)$$~~

$$P(M=1) = P(X=1, Y=1) + P(X=0, Y=1) = 0.1$$

$$P(M=2) = P(X=2)P(Y=1) + P(X=2)P(Y=2) + P(Y=2)P(X=1) + P(Y=2)P(X=0) \\ = 0.1 + 0.2 + 0.12 + 0.08 = 0.5$$

$$P(M=3) = 0.4$$

M 1 2 3

P 0.1 0.5 0.4

0

~~$$P(N=1) = P(Y=1)P(X \leq 1) = 0.2$$~~

$$P(N=1) = 0.3 \times 1 + 0.2 \times 0.5 = 0.4$$

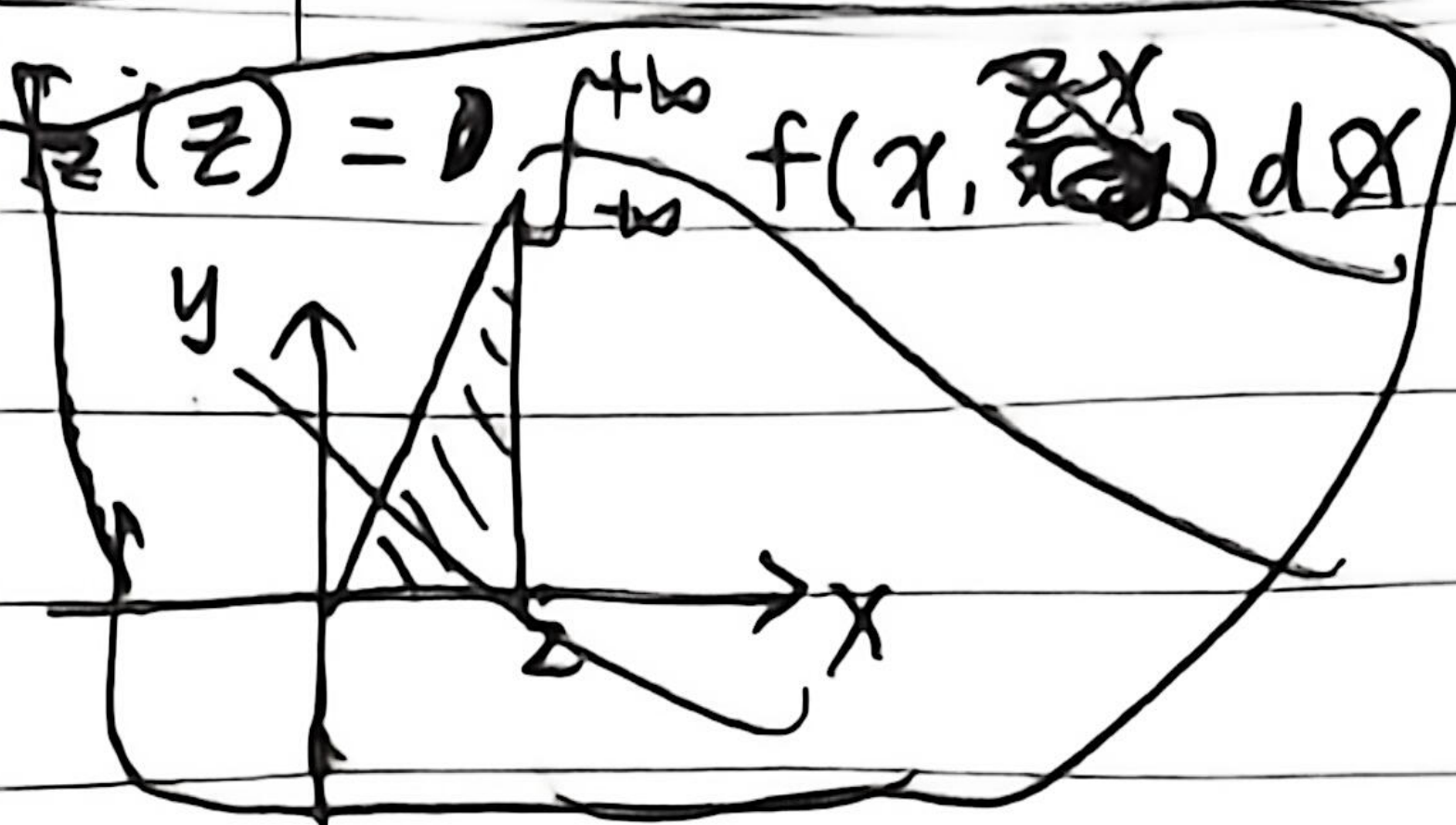
$$P(N=2) = 0.5 \times 0.4 = 0.2$$

N 0 1 2

P 0.2 0.2 0.2

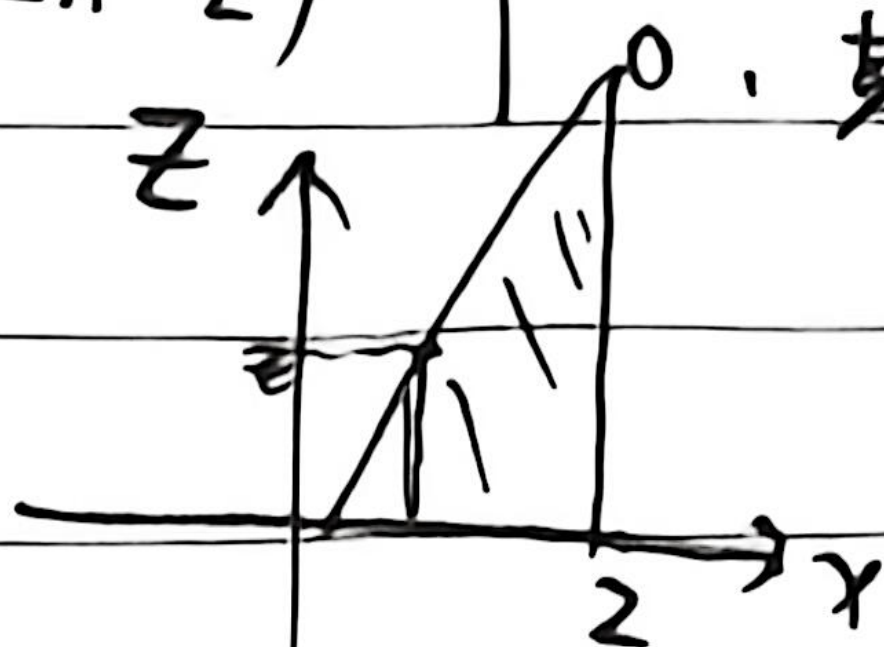


A 36



$$0 < z < 2x$$

$$f(x, 2x-z) = \frac{1}{4x}, \quad 0 < x < 2, \quad 0 < 2x-z < 2x$$



$$f_z(z) = \int_0^2 f(x, 2x-z) dx$$

$$\text{当 } 0 < z < 4, \quad f_z(z) = \int_{\frac{z}{2}}^2 \frac{1}{4x} dx = \frac{2-z}{4} = \frac{1}{2} - \frac{z}{8}$$

$$\text{其他, } f_z(z) = 0$$





A38

$$\int_0^y 2y dy$$



$$F_m(m) = P(M \leq m) = P(X \leq m) P(Y \leq m)$$

$$f_x(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{其他} \end{cases} \quad f_x(x) = \begin{cases} x, & 0 < x < 1 \\ 0, & \text{其他} \end{cases} \quad f_y(y) = \begin{cases} y^2, & 0 < y < 1 \\ 0, & \text{其他} \end{cases}$$

① 若  $0 < m < 1$ ,  $F_m(m) = F_x(m) \cdot F_y(m) = m^3$   
 $m$  其他,  $0$

$$f_m(m) = \frac{\partial F_m}{\partial x \partial y} = \begin{cases} 2y, & 0 < m < 1 \\ 0, & \text{其他} \end{cases}$$

$$f_m(m) = F'_m(m) = \begin{cases} 3m^2, & 0 < m < 1 \\ 0, & \text{其他} \end{cases}$$

② 若  $0 < n < 1$ ,  $F_N(n) = P(N < n) = 1 - P(N > n) = 1 - P(X > n) P(Y > n)$   
 $= 1 - (1 - F_x(n))(1 - F_y(n))$   
 $= 1 - (1 - n)(1 - n^2) = n^2 + n - n^3$

$$f_n(n) = \begin{cases} 2n + 1 - 3n^2, & 0 < n < 1 \\ 0, & \text{其他} \end{cases}$$



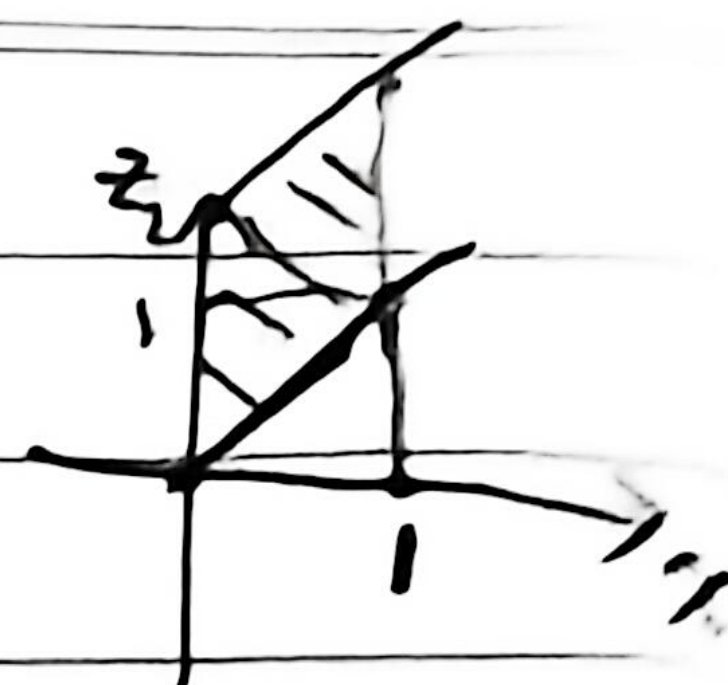


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$$0 < z-x < 2 \quad (1 - \frac{z}{3})$$

$$f_z(z) = \int_0^{\infty} f(x, z-x) dx$$

$$f(x, z-x) = \begin{cases} \frac{3-z}{3}, & 0 < x < 1, x < z < z+x \\ 0, & \text{其他} \end{cases}$$



$$\text{for } 0 < z < 1: f_z(z) = \int_0^z \frac{3-z}{3} dx = (1 - \frac{z}{3})z$$

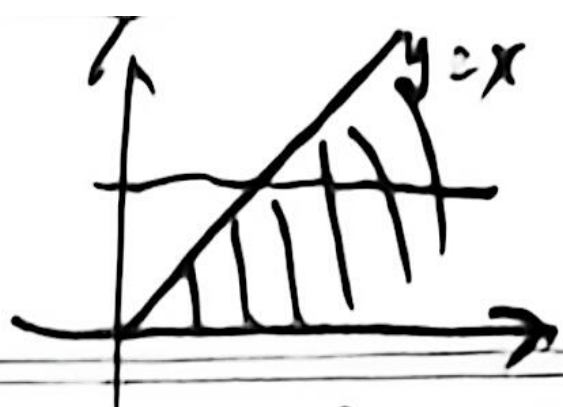
$$1 < z < 2: f_z(z) = \int_0^1 \frac{3-z}{3} dx = 1 - \frac{z}{3}$$

$$2 < z < 3: f_z(z) = \int_{z-2}^1 \frac{3-z}{3} dx = (1 - \frac{z}{3})(\text{crossed out})$$





A2



$$-ce^{-x}$$

$$(1) I = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy = \int_0^{\infty} \int_y^{\infty} ce^{-x} dx dy = \int_0^{\infty} ce^{-y} dy = C = 1$$

$$(2) f_Y(y) = \int_{-\infty}^{\infty} f(x,y) dx = \int_y^{\infty} e^{-x} dx = e^{-y} \quad \text{当 } y > 0$$

$$\therefore f_Y(y) = \begin{cases} e^{-y}, & y > 0 \\ 0, & y \leq 0 \end{cases}$$

$$(3) f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)}$$

$$ye^{-x}|_x$$

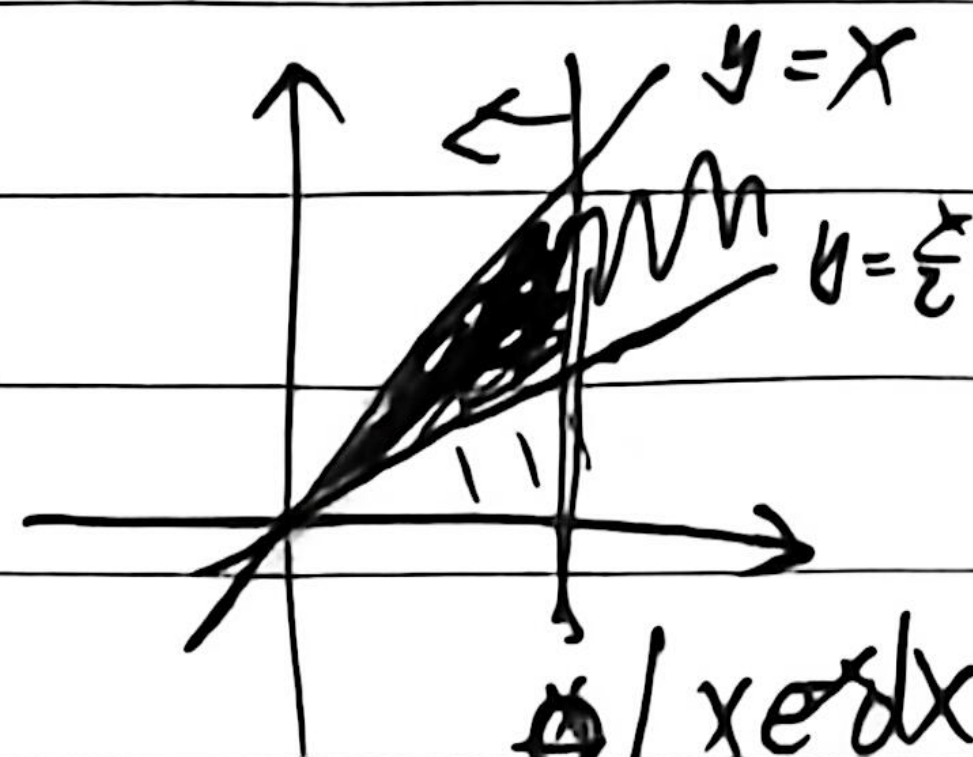
$$\text{其中 } f_X(x) = \int_{-\infty}^{\infty} f(x,y) dy \quad \text{当 } x > 0 \quad f_X(x) = \int_0^x e^{-x} dy = xe^{-x}$$

$$\therefore f_{Y|X}(y|x) = \begin{cases} \frac{1}{x}, & 0 < y < x \\ 0, & \text{其他} \end{cases}$$

$$f_{Y|Z}(y) = \begin{cases} \frac{1}{3}, & 0 < y < 3 \\ 0, & \text{其他} \end{cases}$$

$$P = \int_{-0.2}^2 f_{Y|Z}(y) dy = \int_0^2 \frac{1}{3} dy = \frac{2}{3}$$

$$(4) Z = \begin{cases} X, & Y > X/2 \\ 0, & \text{其他} \end{cases}$$



$$\text{当 } Y \leq \frac{X}{2} \quad F_Z(0) = P(Y \leq \frac{X}{2}) = \frac{1}{2}$$

$$\text{当 } Y > \frac{X}{2} \quad \frac{1}{2} = \int_0^{\infty} dx \int_{\frac{x}{2}}^x e^{-x} dy = \int_0^{\infty} \frac{x}{2} e^{-x} dx = \frac{1}{2}$$

$$(2) F_Z(z) = P(Z < z) = P(X < z, Y > \frac{X}{2})$$

$$= \int_0^z dx \int_{\frac{x}{2}}^x e^{-x} dy = \int_0^z (e^{-x} - e^{-x/2}) dx$$

$$= -e^{-x} + 2e^{-x/2} \Big|_0^z = -ze^{-z} + 2e^{-z/2} + 1$$

$$F_Z(z) = P(Z < z) = P(X < z, Y > \frac{X}{2}) + P(0 < z, Y \leq \frac{X}{2})$$

$$= \frac{1}{2} [1 - (z+1)e^{-z}] + \frac{1}{2}$$





A2)

$$E(x) = \int_1^3 x \cdot \frac{x}{4} dx = \frac{13}{6}$$

$$P\{X > E(X)\} = P\{X > \frac{13}{6}\} = \int_{\frac{13}{6}}^3 \frac{x}{4} dx = \frac{155}{288}$$

A4)

|    |     |     |     |     |
|----|-----|-----|-----|-----|
| XY | 0   | 1   | 2   | 4   |
| P  | 0.4 | 0.1 | 0.3 | 0.2 |

$$E(XY) = 0.1 + 0.6 + 0.8 = 1.5$$

|             |     |     |
|-------------|-----|-----|
| ② min(X, Y) | 1   | 2   |
| P           | 0.3 | 0.7 |

$$E\{\max(X, Y)\} = 0.3 + 1.4 = 1.7$$

③ ~~max(X, Y)~~

|           |     |     |     |
|-----------|-----|-----|-----|
| min(X, Y) | 0   | 1   | 2   |
| P         | 0.4 | 0.4 | 0.2 |

$$E\{\min(X, Y)\} = 0.4 + 0.4 = 0.8$$



2)

|   |   |                |                |                |                |                |                |                |
|---|---|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| X | 1 | 2              | 3              | 4              | 5              | 6              | 7              | 8              |
| P | 0 | $\frac{1}{28}$ | $\frac{2}{28}$ | $\frac{3}{28}$ | $\frac{4}{28}$ | $\frac{5}{28}$ | $\frac{6}{28}$ | $\frac{7}{28}$ |

$$\therefore E(X) = \frac{2}{28} + \frac{6}{28} + \frac{12}{28} + \frac{20}{28} + \frac{30}{28} + \frac{42}{28} + \frac{56}{28} = \frac{168}{28} = 6$$

3)

$$P(X=k) = 2\left(\frac{1}{2}\right)^k = \left(\frac{1}{2}\right)^{k-1}$$

$$\therefore E = \sum_{k=1}^{\infty} P(X=k) \cdot k = \sum_{k=1}^{\infty} k \cdot \left(\frac{1}{2}\right)^{k-1}$$

其中  $\sum_{k=1}^{\infty} k \left(\frac{1}{2}\right)^{k-1} = \frac{1}{(1-\frac{1}{2})^2} = 4$

$$\sum_{k=2}^{\infty} k \left(\frac{1}{2}\right)^{k-1} = 3$$

$$\therefore E = 3$$

